

RAY-VASC PAPER EXECUTION PROMPTS v2.0

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Each prompt below is a complete standalone instruction for generating one paper. Feed the prompt into a session with all knowledge base files uploaded. Request one section at a time for best output quality.

KNOWLEDGE BASE (Upload to Every Session)

- RAY-VASC Cross-Verified Research Synthesis v3.pdf
- Psilocybin Dihexa PSP Huperzine Ivermectin Publications and Citations.docx
- Ray Confirmed Labs + Cell Rejuvenation + Stack Pathways.docx
- RAY-VASC Case Intelligence Report v4.0.docx
- North Star Build Guide v0.3.docx
- The Brain That Remains v1.1.docx (for Papers 2-7 continuity)

PAPER 1: "The Brain That Remains" — COMPLETE ✓

Status: Publication-ready v1.1. Do not rebuild. Submit or post as preprint.

Framework manuscript establishing Regenerative Loop + APPRE + RAY-VASC architecture. Contains full 3-Phase Plasticity Architecture, mycology convergence map, measurement architecture, claim discipline, governance boundaries, safety/ethics, limitations, 40+ APA 7 citations. Target: Frontiers in Neuroscience or equivalent. Logan Henderson first author, Raymond A. Hornsby acknowledged as human origin (authorization on file).

PAPER 2: RAY-VASC Protocol Manuscript — ACTIVE

Full Title

RAY-VASC: A Prospective N-of-1 Longitudinal Protocol for Tracking Cognitive Motor Mood Respiratory EEG and Biomarker Trajectories Under a Neuroplasticity-Oriented Observational Framework

Your Role

You are writing Paper 2 as Logan Henderson originating researcher. Target journal: JMIR Research Protocols or Frontiers in Neuroscience Methods. Article type: Study Protocol. This document must be externally implementable without Logan present.

What This Paper Is

The protocol paper converts the RAY-VASC framework into an operational study design others can run. It is a 12-month prospective N-of-1 longitudinal observational protocol. It does not report results. It defines the measurement architecture inclusion/exclusion criteria safety monitoring data capture plan analysis approach and governance boundaries so that an outside team can register it on OSF and begin implementation under appropriate ethics governance.

Core Framing (Use Exactly)

The Regenerative Loop + APPRE + RAY-VASC architecture is designed to test whether coordinated daily biological loops and structured adaptive measurement can identify and amplify residual adaptive capacity in complex neurodegenerative and vascular phenotypes. It is not a treatment claim. It is a testable framework.

Manuscript Structure (Write Every Section in Full)

Abstract — Background rationale design outcomes timeline open-source availability

Introduction — Rationale for N-of-1 design in complex multimorbidity. Why standard RCT is premature for hypothesis-generating mechanistic work. Gap: no structured open-source protocol exists for combined neurotrophic mycology plus Regenerative Loop intervention in neurodegenerative phenotypes. Cite FINGER POINTER Bredesen as multidomain precedents.

Objectives and Hypotheses — Primary: feasibility and measurement reliability over 90 days. Secondary: directional signal in cognitive motor mood sleep domains vs baseline. Exploratory: biomarker trajectory vs expected natural history decline rates. Pre-specified hypothesis: A standardized neurotrophic mycology stack embedded within a structured Regenerative Loop protocol will produce measurable directional changes in at least two primary outcome domains within 90 days as detected by a pre-specified 0.3 SD threshold.

Study Design — Prospective longitudinal N-of-1 observational 12-month window. Phase 1 Pre-Load weeks 1-4. Phase 2 Plasticity Window weeks 5-16 only if lawful clinician-governed ethics-approved. Phase 3 Sustain weeks 17-52. Crossover blocks for mycology variation. 0.3 SD signal threshold. Confounder exclusion rules documented.

Eligibility Criteria — Write generically. Inclusion: adult 50+ confirmed neurodegenerative or neurovascular diagnosis caregiver available ability to complete daily logs wearable access physician clearance. Exclusion: active infection within 30 days major medication change within 14 days eGFR below 45 for NAC contraindicated supplements per pharmacist review concurrent interventional trial.

Intervention — Three-Phase Architecture

Phase 1 Pre-Load: Daily standardized neurotrophic mycology stack written generically by compound class: Lion's Mane extract 500-1000mg NGF/BDNF Bacopa Monnieri 300mg AChE/BDNF NAC 600mg GSH precursor anti-tau Cordyceps 500-1000mg mitochondrial support Citicoline 500-1000mg ACh synthesis Reishi 500-1000mg neuroinflammation Omega-3 EPA-DHA 2000mg vascular membrane. AM-PM split per circadian alignment. Anti-tolerance cycling logic documented.

Phase 2 Plasticity Window: ONLY when jurisdictionally legal clinician-governed ethics-IRB-approved. Write as governance placeholder: If and when a licensed clinical oversight arrangement is established in a jurisdiction where such research is lawful a single or limited-session psilocybin-assisted therapy component may be incorporated as an as-administered observational exposure field. This component requires separate IRB or ethics determination. Do not include dosing preparation or administration instructions.

Phase 3 Sustain: Continuation of Phase 1 stack plus Regenerative Loop reinforcement exertion sleep hygiene circadian light social engagement.

Regenerative Loop Protocol — Exertion: 150 min/week moderate intensity plus daily cognitive challenge. Circadian: morning outdoor light minimum 20 minutes within 60 minutes of wake consistent sleep/wake within 30 minutes. Sleep: CPAP for OSA iron correction for PLMD sleep hygiene protocol. Social: minimum 3 meaningful interactions per week caregiver co-participation in daily check-in.

Primary Outcomes — Cognitive: MoCA Trail Making A and B verbal fluency. Motor: TUG 10-meter walk mPSPRS-15 or MDS-UPDRS Part III. Mood: PHQ-9 GAD-7 PCL-5. Sleep: PSQI actigraphy efficiency and latency CPAP AHI. Respiratory: SpO2 6MWT mMRC dyspnea.

Secondary Outcomes — EEG: resting-state spectral power P3 ERP amplitude and latency frontal-central coherence. Biomarkers: p-tau217 plasma BDNF serum CRP IL-6 homocysteine A1c eGFR ferritin Vitamin D.

Data Capture Plan — Daily log 60-90 seconds cognition mood motor sleep breathlessness pain adherence 0-10 scales. Weekly battery TUG SpO2 global impression adherence rate side effect log. Monthly full cognitive battery mood scales respiratory. Quarterly biomarker draws and EEG. Wearable continuous.

Adverse Event Monitoring — Categories mild/moderate/severe. Stopping rules: severe AE requires immediate pause and clinical review. Moderate AE on 2+ consecutive days triggers stack simplification. All AEs logged with date description severity attribution causality resolution.

Statistical Analysis Plan — Time-series visualization all primary outcomes. Baseline minimum 14 clean observation days. Signal threshold 0.3 SD sustained across 2+ consecutive measurement cycles. Confounder-day exclusion sensitivity analysis. Paired t-test or Wilcoxon signed-rank as appropriate. Missing data: 70% adherence minimum for period inclusion.

APPRE Closed-Loop Logic — Subject Profile maps to Inputs maps to Outcomes maps to Updated Response Policy. N-of-1 crossover block design with 0.3 SD threshold. Anti-tolerance cycling. Confounder exclusion rules. One variable at a time tuning. Distinguish APPRE from static

protocol: APPRE learns the individual response curve static protocols assign the same intervention to everyone.

Ethics and Governance — IRB or ethics review required before external prospective data collection. OSF pre-registration precedes data collection. Self-experimentation follows all AE monitoring and stopping rules.

Data Sharing — Protocol documents forms schemas and analysis pipelines publicly available at github.com/onlygreentrades/ray-vasc-regenerative-loop. No individual-level clinical data published without separate consent.

Limitations — N-of-1 limits generalizability. Multiple simultaneous interventions limit causal attribution. Subjective scales susceptible to expectation bias. Controlled-substance layer excluded from open-protocol version and requires separate governance.

Conflict of Interest and Funding — Self-funded. No commercial product or clinical service associated with this protocol.

Acknowledgments — This protocol was motivated by the care of Raymond A. Hornsby. Authorization for attribution is on file.

References — CONSORT N-of-1 extension SPIRIT checklist FINGER POINTER Mori 2009 Moliner 2023 UCSF PD pilot Blest-Hopley 2025 Vohra 2015. Browse and verify all before including.

Claim Discipline Enforced

Never write: repairs brain damage reverses dementia cures PSP proven treatment clinical recommendation. Always write: designed to test may modulate hypothesized to evidence suggests hypothesis-generating observed signal. Flag every unsupported claim as NEEDS CITATION.

Output Rule

Deliverable first. Then constraints and risks. Then QA blocks (Discrepancies Assumptions Open Questions References).

PAPER 3: Ray Case Report — REFRAMED

Full Title

Longitudinal Baseline Characterization in a Complex Neurovascular Parkinsonian Phenotype: Documenting Pre-Intervention Architecture and Measurement Feasibility for Future Neuroplasticity Research

Your Role

You are writing Paper 3 as Logan Henderson originating researcher with Raymond A. Hornsby as the documented case subject. Target journal: *Frontiers in Neurology Case Reports* or *Journal of Medical Case Reports*. Article type: Case Report following CARE checklist. Authorization for full naming and clinical detail is on file from Christine Hood Hornsby POA.

What This Paper Is

This is NOT an intervention outcome paper. Raymond declined to the point of requiring full-time assisted care approximately two weeks before intervention could begin. This paper documents the most richly phenotyped pre-intervention baseline in the literature for this phenotype (PSP vascular dementia Parkinsonism COPD PLMD boxing history familial Parkinsonism thalamic infarcts white matter disease). It demonstrates what data were collected what data were not collected why and uses this case to show why earlier intervention windows matter. It is hypothesis-generating and contributes the baseline measurement architecture. This is publishable. The absence of intervention data is part of the scientific record not a failure.

Manuscript Structure (Write Every Section in Full)

Abstract — Background: Complex neurodegenerative phenotypes with mixed PSP vascular and traumatic injury burden are poorly characterized at high temporal resolution. Case presentation: 66-year-old male with PSP VCID Parkinsonism COPD PLMD boxing history familial risk richly phenotyped with neurology MRI DaTSCAN labs sleep study EEG baseline. Conclusion: This case documents feasibility of comprehensive pre-intervention baseline measurement and demonstrates the clinical and temporal constraints that determine intervention windows in advanced disease.

Introduction — PSP and vascular cognitive impairment are rarely studied with combined neurophysiological metabolic sleep respiratory and detailed family-history phenotyping. This case represents an opportunity to document a complete pre-intervention architecture before disease progression and care-transition constraints made intervention unfeasible. The contribution is methodological: it demonstrates what can be measured and what barriers exist.

Case Presentation — Timeline — Feb 2026: Neurology dx PSP MoCA 11/30 UPDRS Motor 52 Hoehn-Yahr Stage 3. Jan 2026: DaTSCAN severely reduced striatal uptake DaTQUANT unable to quantify. Feb 2026: MRI Brain advanced confluent white matter disease bilateral thalamic and right cerebellar remote infarcts NeuroQuant Hippocampal Occupancy Score 0.6 1st percentile hippocampal volume 4.7cc 11th percentile inferior lateral ventricles 3.3cc 98th percentile. March 2026: Labs eGFR improved 65 to 81-83 CRP 0.62 optimal TC 92 LDL 27 critically low on rosuvastatin HbA1c 5.7 borderline homocysteine 12 $\mu\text{mol/L}$ folate near-ceiling. May 2025: Sleep study PLMI 137.66/hr sleep efficiency 39.48% AHI resolved to 0 on CPAP at 11 cmH₂O CPAP intolerance confirmed. Family history: Parkinsonism in father and brother possible MAPT variant. Boxing history age 13-19/20 repetitive head impacts during PFC maturation window.

What Was Not Collected — EEG baseline IL-6 TNF- α BDNF panel p-tau217 NfL intervention-phase data. The absence of these biomarkers limits the case report to hypothesis-generating

observational documentation which is exactly what this paper is.

Clinical Course and Care Transition — By late March 2026 Raymond required full-time assisted care. Intervention window was lost approximately two weeks before readiness. This is documented as a timing constraint not a protocol failure.

Discussion — Why This Case Matters — Demonstrates feasibility of comprehensive phenotyping in complex mixed neurodegenerative phenotype. Documents the real-world clinical and family constraints that determine intervention windows. Highlights the need for earlier detection and faster clinical decision pathways. Contributes measurement architecture that future teams can use.

Patient Perspective — Include brief statement from Christine Hood Hornsby on Ray's perspective and family motivation for contributing to research. Authorization on file.

Informed Consent — Written authorization for public naming and clinical data publication provided by Christine Hood Hornsby Power of Attorney for Raymond A. Hornsby.

Acknowledgments — The family's willingness to contribute Ray's clinical story to the research record honors his life and may help future patients and families facing similar diagnoses.

References — Cite PSP diagnostic criteria vascular dementia criteria DaTSCAN interpretation NeuroQuant validation sleep and PLMD literature thalamic infarct and cognition MAPT variants TBI and neurodegeneration.

Claim Discipline

This is observational only. No intervention claims. No outcome claims. Honest documentation of what was and was not collected and why.

PAPER 4: APPRE Data Architecture — QUEUED

Full Title

APPRE: An Open Data Architecture for Adaptive N-of-1 Neuroplasticity Protocols

Your Role

You are writing Paper 4 as Logan Henderson originating researcher. Target: JMIR Medical Informatics or Frontiers in Digital Health. Article type: Technical Report Methods.

What This Paper Is

Publishes the reusable data system schema set and analysis pipeline so external teams can implement data capture immediately. APPRE Subject Profile Inputs Outcomes Updated Response Policy. Includes schema definitions data dictionaries validation rules confounder annotation adverse event structure and closed-loop tuning logic.

Manuscript Structure

Abstract Introduction (gap: no open adaptive N-of-1 data standard exists for neuroplasticity) APPRE Architecture (diagram and logic flow) Schema Specifications (JSON or YAML with field definitions) Data Capture Workflow (daily weekly monthly quarterly) Confounder Annotation Rules Adverse Event Structure Analysis Pipeline (time-series signal detection 0.3 SD threshold) Implementation Guide (how external team deploys APPRE) Limitations Discussion References.

Output Must Include

Complete machine-readable schema files as supplementary materials or GitHub links.
Synthetic example data demonstrating schema use. Reproducibility checklist.

PAPER 5: Regenerative Loop Theory — QUEUED

Full Title

The Regenerative Loop: A Multidomain Reinforcement Model for Neuroplasticity Enabling Environments

Your Role

You are writing Paper 5 as Logan Henderson originating researcher. Target: *Frontiers in Neuroscience Hypothesis and Theory* or *Neuroscience and Biobehavioral Reviews*. Article type: Theory Framework.

What This Paper Is

Formalizes the exertion sleep circadian social-safety architecture as a standalone peer-reviewed contribution. Each pillar gets mechanism evidence grade biological rationale and interaction dynamics. This is the theoretical spine for why multidomain daily loops matter for neuroplasticity outcomes.

Manuscript Structure

Abstract Introduction (single-factor models fail multidomain precedents FINGER POINTER) Four Pillars Detailed (Exertion as BDNF driver Sleep as glymphatic repair Circadian as temporal gate Social safety as HPA buffer and adherence anchor) Loop Dynamics (how pillars amplify or constrain each other) Evidence Grading (A for exercise and sleep B for circadian and social) Measurement Architecture (how to operationalize each pillar) Discussion Limitations References.

PAPER 6: Mycology Evidence Map — QUEUED

Full Title

Evidence Map for Neurotrophic Mycology and Neuroprotective Compound Stacks in Vascular Neurodegeneration: A Graded Systematic Review

Your Role

You are writing Paper 6 as Logan Henderson originating researcher. Target: Nutrients or Frontiers in Nutrition. Article type: Systematic Review Scoping Review.

What This Paper Is

The compound evidence library becomes a standalone publication. Graded A-E. Documents the state of knowledge and gaps. Directly supports APPRE compound selection philosophy.

Manuscript Structure

Abstract Introduction (gap: no graded synthesis exists for combined mycology neurotrophic stacks) Methods (search strategy inclusion/exclusion evidence grading A-E) Results (evidence tables for Lion's Mane Bacopa NAC Cordyceps Citicoline Reishi Omega-3 psilocybin preclinical and clinical) Convergence Hypothesis (all converge on BDNF-TrkB-mTOR synaptogenesis) Discussion (what is known what remains unknown) Limitations References.

PAPER 7: 3-Phase Plasticity Architecture — QUEUED

Full Title

Three-Phase Plasticity Architecture: A Framework for Structuring Pre-Load Plasticity Window and Sustain/Integration Phases in Neuroplasticity Research Protocols

Your Role

You are writing Paper 7 as Logan Henderson originating researcher. Target: Frontiers in Neuroscience Methods or Journal of Neuroscience Methods. Article type: Framework Methods.

What This Paper Is

Formalizes the 3-phase system as a reproducible design contribution to the field. Pre-Load as stabilization and neurotrophic readiness. Plasticity Window as governed mechanistic intervention layer. Sustain as durability testing and regression detection.

Manuscript Structure

Abstract Introduction (gap: no staged architecture exists for testing coordinated neuroplasticity interventions) Phase 1 Pre-Load Definition (purpose measurement safety gates) Phase 2 Plasticity Window Definition (governance boundaries controlled-substance framing) Phase 3 Sustain Definition (durability regression detection) Staged Hypothesis Testing Logic (Phase 1 testable independently Phase 2 optional Phase 3 tests persistence) Measurement Windows and Gates Discussion Limitations References.

Not medical advice. Not legal advice. Research manuscript execution prompts only.

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